

AVVARI SWETHAN KARTHIK

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PROFESSIONAL SUMMARY

Computer Science (AI & ML) student with industry exposure through an Accenture internship and practical experience in software development and AI-driven applications. Proficient in Python, PyTorch, and modern generative AI frameworks. Driven to design scalable backend systems, solve complex engineering challenges, and build technology solutions with real-world impact.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science and Engineering (AI & ML) 2023 – 2027 | Pothavarappadu, Andhra Pradesh
NRI Institute of Technology, Pothavarappadu, Andhra Pradesh
Aggregate Score: 89% (Cumulative up to 3rd Year, 2nd Semester)

PROFESSIONAL EXPERIENCE

Accenture Summer Internship May 2026 – Jul 2026 | Hyderabad, India
Associate Software Engineer Intern

- Developed agentic AI solutions for text summarization, FAQ assistance, meeting analysis, and multi-agent research workflows.
- Built scalable serverless REST APIs using Python, Azure Functions, Azure OpenAI, and structured backend architecture.
- Gained hands-on experience in prompt engineering, cloud AI application development, API integration, and agile software engineering.

Infosys Springboard 6.0 Aug 2025 – Sep 2025 | Remote
Artificial Intelligence Intern

- Built "Mood Music AI," an intelligent emotion-based music generation system.
- Implemented NLP and Generative AI pipelines in Python to seamlessly map text to audio.
- Mastered end-to-end AI product development from theoretical modeling to functional deployment.

Coincent.ai (IIT Madras e-Cell) Aug 2024 – Oct 2024 | Remote
Artificial Intelligence Intern

- Developed a predictive machine learning model focused on automated handwriting analysis.
- Executed end-to-end data preprocessing, model training, and hyperparameter tuning in Python.
- Strengthened foundational ML engineering, model evaluation, and data-driven problem-solving skills.

PROJECTS

AI-Based Music Composition System

- Engineered a complete text-to-audio generation web application utilizing Python, PyTorch, Hugging Face models, and Streamlit to output mood-aligned music tracks.
- Addressed the manual, time-consuming process content creators face when sourcing royalty-free, emotion-specific background audio.
- Implemented a seamless pipeline combining BART Large MNLI for mood classification and MusicGen to instantly automate custom music generation.

Pothole Detection and Reporting System

- Developed a high-performance computer vision backend utilizing YOLOv8, FastAPI, and Python to process visual video feeds and extract GPS-tagged road damage data.
- Targeted the inefficiency of manual municipal road inspections that lead to delayed hazard reporting and maintenance.
- Automated the hazard identification process by routing structured detection data through RESTful APIs, facilitating rapid and accurate municipal reporting.

NutriSync: AI-Powered Personalized Dietitian System

- Built an automated, API-driven diet planning engine in Python, integrating PDF generation libraries to dynamically calculate complex caloric and macronutrient algorithms.
- Solved the complexity individuals face when manually calculating macronutrients and translating raw health metrics into actionable dietary goals.
- Streamlined nutritional tracking by building a dynamic backend system that instantly processes user parameters to serve customized, downloadable diet plan reports.

TECHNICAL SKILLS

Programming Languages Python, Java, C	AI / ML & Frameworks PyTorch, YOLOv8, Hugging Face Transformers, NLP, Computer Vision, Generative AI	Web & Backend Development FastAPI, Streamlit, HTML, CSS, JavaScript, React	Database MongoDB, Firebase, SQL Server
Tools Git, Github, VS Code, Jupyter Notebook			

CERTIFICATIONS & ACHIEVEMENTS

Certifications
Salesforce Agentforce Specialist, HackerRank Python, 15+ Infosys Springboard AI Specializations.

Hackathons
Developed "Project Kisan" for Google Cloud Agentic AI Hackathon; competed in Adobe India, MindSprint, and HackVibe 2k25.